

The radar-based system provides depth perception and can 'see' through soft materials, such as a blanket covering a child in a child restraint.

Tesla adds, "It can differentiate between a child and an object left on the seat, reducing the likelihood of false alarms, and can detect micromovements like breathing patterns and heart rates, neither of which can be captured by cameras or

in-seat sensors alone."

Radar imaging can assess body size to optimise airbag deployment in a crash depending on whether an adult or child is seated, which it says would be more effective than existing weight-based and in-seat sensor systems. It would also more accurately determine when to engage seat belt reminders.

Government launches contest to promote homegrown microprocessor companies

The government of India has launched a contest to promote homegrown microprocessor companies in line with its aim of Atmanirbhar Bharat. This comes after indigenously developed microprocessors 'Shakti' and 'Vega' have been rolled out under the Microprocessor Development Programme of the Ministry of Electronics and IT.

Under the Swadeshi Microprocessor Challenge, hundred shortlisted companies will win a grant of total ten million rupees, while 25 finalists will win ten million rupees each. The top ten teams entering the finale will get the seed fund

of total 23 million rupees and twelve months of incubation support.

Launching the initiative, Ravi Shankar Prasad, Union Minister of Law & Justice, Communications and Electronics & Information Technology, said that Swadeshi Microprocessor Challenge - Innovate Solutions for Aatmanirbhar Bharat will provide further impetus to the strong ecosystem of startup, innovation, and research in the country. The government has already launched content to promote a homegrown videoconferencing tool along with giving a push to local mobile applications.

This hands-free face shield will let you take calls and listen to music

Indian helmet manufacturer Steelbird has launched a unique hands-free face shield called IGN-1 HF Static Face Shield. Users can connect the face shield to their smartphones with the help of a cable. The face shield has an inbuilt speaker, which can

double up as an over the ear headphone, helping you access phone calls and also listen to music.

The company mentions that its hands-free shield will help users avoid contact with their smartphone, which is one of the carriers of Covid-19.

The face shield does not need a battery and is compatible with all the mobile phones. The device comes with adjustable ear pads to suit any individual's ear size and position. Further, shield claims to offer high-quality sound and noise cancellation through a single directional microphone. The fact that it is IP5 water resistant makes it adequately waterproof as well.

"With the widespread coronavirus, we thought of coming with a product that gives 360-degree protection to people. So, we innovated IGN-1 HF Static Face Shield, which is a first of its kind revolutionary hands-free face shield. This face shield will not only help you to save yourself from coronavirus infection, but you can also pick or receive a phone call and speak to the caller through the inbuilt battery-less speakers and microphone system. One can also enjoy music through the Face Shield," says Rajeev Kapur, managing director, Steelbird Group.



Steelbird's unique hands-free face shield (Credit: Amazon)

IIT Roorkee launches online programs on AI and data science

The Indian Institute of Technology, Roorkee has announced a partnership with an online learning platform, Coursera. As a part of this partnership, the two organisations have launched two new certificate programs for learners around the world. While the first program is focused on artificial intelligence (AI) and machine learning (ML), the other one focuses on data science. Both these programs will be available on Coursera in early 2021.

Talking about the programs, the six-month certificate program in AI and ML will consist of video lectures, hands-on learning oppor-

tunities, team projects, tutorials, and workshops. This program will also teach classical ML techniques and provide hands-on programming experience with Tensorflow for model building, robust ML production, and powerful experimentation to the learners.

The certificate program in data science, on the other hand, will equip professionals looking to build skills in data science, ML, critical thinking, data collection, data visualisation, and data management. The program will help develop Python and SQL programming skills necessary for a career in data analytics.